THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The point on the y-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

2. The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

(a) 64 **(b)** 65 **(c)** 62 **(d)** 63

3. If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

(a) $\frac{13}{3}$ **(b)** $\frac{16}{3}$ **(c)** $\frac{19}{3}$ **(d)** $\frac{22}{3}$

4. The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the y-axis is 3: 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

5. The side of the square with diagonal endpoints $(0, 0)$ and $(6, 8)$ has squared length: 1

(a) 49 **(b)** 52 **(c)** 50 **(d)** 51

6. The three points $(0, 0)$, $(8, 0)$ and $(3, 7)$ enclose area: 1

(a) 27 **(b)** 30 **(c)** 29 **(d)** 28

7. The point on the y-axis equidistant from $(1, 2)$ and $(5, 4)$ has $y =$ 1

(a) 9 **(b)** 8 **(c)** 10 **(d)** 11

8. The area of the triangle formed by $(11, 0)$, $(0, 4)$ and the origin, doubled, is: 1

(a) 45 **(b)** 44 **(c)** 46 **(d)** 43

9. If $P(x, y)$ is equidistant from $(2, 4)$ and $(8, 2)$ and lies on $y = 0$, then $x =$ 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

10. The ratio in which $(-2, 4)$ and $(8, 1)$ are joined by the y-axis is 2: 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that the points $A(0, 0)$, $B(3, 4)$, $C(8, 4)$, $D(5, 0)$ are the vertices of a rhombus. 2

12. Find the value of k for which the points $(1, 5)$, $(2, 8)$ and $(4, k)$ are collinear. 2

13. The vertices of a triangle are $(0, 0)$, $(6, 0)$, $(0, 8)$. Find the length of the median from the first vertex. 2

14. Show that the points $A(1, 2)$, $B(4, 6)$, $C(9, 6)$, $D(6, 2)$ are the vertices of a rhombus. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the points $A(1,\ 1)$, $B(5,\ 4)$, $C(10,\ 4)$, $D(6,\ 1)$ are the vertices of a rhombus. 3

16. The line joining $(0,\ 0)$ and $(9,\ 6)$ is trisected by points P and Q. Find their coordinates. 3

17. Find the values of y for which the distance between $P(8,\ y)$ and $Q(2,\ 3)$ is 10 units. 3

OR

Show that the points $A(0,\ 0)$, $B(6,\ 8)$, $C(16,\ 8)$, $D(10,\ 0)$ are the vertices of a rhombus. 3

18. The line joining $(3,\ 3)$ and $(12,\ 9)$ is trisected by points P and Q. Find their coordinates. 3

OR

Find the values of y for which the distance between $P(17,\ y)$ and $Q(5,\ 1)$ is 13 units. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The vertices of $\triangle ABC$ are $A(0,\ 0)$, $B(12,\ 0)$ and $C(6,\ 8)$. Find the coordinates of the centroid and the area of the triangle. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(-2,\ 1)$, $B(1,\ 0)$ and $C(4,\ 3)$. Find the fourth vertex D. 5

20. The vertices of $\triangle ABC$ are $A(1,\ 2)$, $B(7,\ 2)$ and $C(4,\ 8)$. Find the coordinates of the centroid and the area of the triangle. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(1,\ 5)$, $B(7,\ 3)$ and $C(10,\ 9)$. Find the fourth vertex D. 5

END OF QUESTION PAPER — Total: 40 Marks

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